

An infinite prime chain inside a localization of a PI ring

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Disproof of conjecture 00000009976. The conjecture asserts finiteness of prime localization chains, with a bound polynomial in the polynomial identity degree. A commutative polynomial ring provides an actual infinite chain in a single localization, with fixed identity degree.

Let

$$R = \mathbb{Q}[X_0, X_1, X_2, \dots].$$

This ring is commutative, so it satisfies the nonzero polynomial identity $UV - VU = 0$ of degree two in the free associative algebra. Thus it is a PI ring. If “PI degree” denotes the conventional PI degree of a commutative domain, that degree is one instead; either convention gives a fixed finite degree here.

For each $n \geq 0$, let $\kappa_n : R \rightarrow R$ be the substitution

$$\kappa_n(X_i) = \begin{cases} 0 & i < n, \\ X_i & i \geq n, \end{cases} \quad P_n = \ker \kappa_n.$$

Because the target ring is an integral domain, P_n is prime. Equivalently $P_n = (X_0, \dots, X_{n-1})$, and its quotient is the polynomial ring in the remaining variables. If $m \leq n$ then $\kappa_n \kappa_m = \kappa_n$, hence $P_m \subseteq P_n$. Moreover

$$X_n \in P_{n+1}, \quad X_n \notin P_n.$$

Thus $P_0 \subsetneq P_1 \subsetneq P_2 \subsetneq \dots$ is an infinite strict chain of prime ideals.

Let $\varepsilon : R \rightarrow \mathbb{Q}$ evaluate every X_i at zero, and let $M = \ker \varepsilon$. It is prime (indeed maximal, since ε is surjective), and $\varepsilon \kappa_n = \varepsilon$ shows that every $P_n \subseteq M$.

Now use the actual localization $R_M = (R \setminus M)^{-1}R$. Prime localization correspondence is the order isomorphism

$$\{\text{prime ideals of } R_M\} \cong \{\text{prime ideals of } R \text{ contained in } M\}.$$

Its inverse extends an ideal to R_M , and contraction recovers the original prime ideal. Consequently $Q_n = P_n R_M$ is prime and

$$Q_0 \subsetneq Q_1 \subsetneq Q_2 \subsetneq \dots$$

For completeness, strictness cannot disappear: if $Q_n = Q_{n+1}$, contracting both sides would give $P_n = P_{n+1}$, contradicting the variable witness above. This is an infinite prime chain inside a single localization of a ring with fixed PI degree. In particular there are finite initial chains of every length, so no finite polynomial bound in that degree can hold.

The asserted chain finiteness is therefore false. We need not dispute the separate spectral-space claim or assign a meaning to the source’s unspecified comparison topology: its explicit finiteness clause already fails. A Noetherian or finite-dimensional restriction would be additional to the bilingual source, which imposes neither.

Lean coverage. The project uses the actual ring of multivariate polynomials in countably many variables over $\mathbb{Q}$. It constructs substitution homomorphisms, their prime kernels, containment in the same prime M , and strictness from actual variables. Crucially it then uses the standard library's prime order isomorphism for the actual ring `Localization.AtPrime M`, producing a formal infinite strict chain of localized prime ideals. It also proves the fixed degree-two commutator identity. Lean 4.19.0 and Mathlib are publicly pinned; printed theorem audits use only standard logical axioms.